Data Wrangling – Problem Solving (SQL)

(Teaching Notes – Step-by-step Thought Process)

Initial Setup

1. Open **SSMS**
2. Login with credentials
3. Go to database: msk1-masterclass

Q1. Find all customers living in California

Step 1: First Guess

select * from mskl_customers

where state ='california';

Step 2: Check actual values

select distinct(state)

from mskl_customers;

What we notice

Multiple representations of California:

- "California"
- "calfiornia"
- " california"
- "California "

➡️ **This is a data quality issue**

- Spelling
- White space

Step 3: What function should we use?

We need:

- To match variations → LIKE
- To remove white space → TRIM()

Step 5: Correct Query

```
select *
from mskl_customers
where trim(state) like 'cal%ornia';
```

Why this works

- TRIM() removes extra spaces
- LIKE handles spelling/case variations
- Covers **all instances of California**

Q2. Customers in Florida or New York AND in Consumer segment

Step 1: Look at the data

```
SELECT *
FROM msk1_customers;
```

Step 2: First attempt (WRONG)

```
SELECT *
FROM mskl_customers
WHERE state = 'florida' OR  state = 'new york'
  AND segment = 'consumer';
```

Because if we notice at the data we can see all the segments are showing

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Step 3: Fix using parentheses

```
select * from mskl_customers
where (state = 'florida' or state = 'new york')
and segment = 'consumer';
```

Step 4: Cleaner approach using IN

```
select * from mskl_customers
where state in('florida','new york')
and segment = 'consumer';
```

Q3. Print customer names in format

Input: ASHISH SHAH
Output: A. SHAH

Step 1: Understand the problem visually

Show in whiteboard

ASHISH SHAH

Step 2: Excel logic (analogy)

- First letter → LEFT() + ' . '
- Space position = 7
- Total length = 11
- Last name → RIGHT(name,len-space)

Step 3: SQL equivalent functions

- LEFT() for first part
- SUBSTRING(name, from where , how many)
- LEN() total length
- CHARINDEX() to get position of space

Step 4: SQL Query

```
select * ,

left(customer_name,1) + ' . ' +

substring( customer_name, charindex(' ', customer_name)+ 1,len(customer_name)-charindex(' ', customer_name))

from mskl_customers
```

Note

- Can also be solved using REVERSE() (try later)

Q4. How many customers have an Outlook email?

Step 1: Identify table

- Table: msk1_customers

So we can like operator for spelling issue
we can use like and % to deal with it

Step 2: Pattern matching

```
select *

from customers

Where customer_email like '%@outlook.com';
```

To just get the count

```
select count(*)

from customers

Where customer_email like '%@outlook.com'
```

Q5. Orders placed in 2018

Step 1: Identify column

From orders table

Select * from orders;

- order_purchase_date

Step 2: Counting orders

SELECT COUNT(*)

FROM orders

WHERE YEAR(order_purchase_date) = ’2018’;

Step 4: Duplicate customer issue

Same customer may place multiple orders

Step 5: Better query

SELECT COUNT(DISTINCT customer_id)

FROM orders

WHERE DATEPART(year, order_purchase_date) = 2018;

Alternative functions
using datepart

select count(distinct customer_id) from orders where datepart(year,order_purchase_date) = '2018'

using datename

select count(distinct customer_id) from orders where datename(year,order_purchase_date) = '2018'

we will see difference with the help of next qn

Q6. Customers in December 2017

Using YEAR + MONTH

```
select count(distinct customer_id)
from orders
where year(order_purchase_date) = '2017'
and month(order_purchase_date)= 12;
```

Using DATEPART

```
select count(distinct customer_id)
from orders
where datepart(year,order_purchase_date) = 2017 and
datepart(month,order_purchase_date)= 12;
```

Using DATENAME (string-based)

```
select count(distinct customer_id)
from orders
where datename(year,order_purchase_date) = 2017 and
datename(month,order_purchase_date)= 'december';
```

Key Teaching Point

Function	Output
YEAR	Integer
DATEPART	Integer
DATENAME	String

Q7. Weekday logic (calendar understanding)
Hint:dateadd and weekday

Select * from orders;

Explain the qn and the example given in the qn

Explain with todays date what might be upper limit and lower limit

in sql sun will be 1 and sat will be 7

then chose a date which lies In middle of the week

From the middle date how can we jump to lower limit(sun) and upper limit(sat)

BY adding dates and removing dates we can
+ 3 days and -3 days from middle days
so all caluclations are done with respect to 7

show todays example

so first we need to find out the weekday

ex
select datepart(weekday,'2026-01-18')

select datepart(weekday,'2026-01-24')

now to add or remove days we can use dateadd
dateadd(what,how many,where)

what tells us whether to add days,hours,week ……

so using the dateadd we jump to lower limit and upper limit
BY using the number 7 with respect to weekday

Sunday = 1
Saturday = 7

so we will use

so in order to get the upper we need to add

7 - dateadd(weekday, order_purchase_date)

example
7 – middle date
= 3 days

to get lower limt

- dateadd(weekday, order_purchase_date)

-4 days
We need to get -3 not -4

so
- dateadd(~~weekday~~, order_purchase_date) + 1
Day

dateadd -> we will use for both addition & Sub

select order_purchase_date , -> Day

dateadd(~~weekday~~, - datepart(weekday,order_purchase_date)+1,order_purchase_date) as lower,
dateadd(~~weekday~~, 7 - datepart(weekday,order_purchase_date),order_purchase_date) as uppper

from orders; -> Day

Now we need to get in that format
6th may to 12th may

for that we can use format

format(what needs to be format, Into what) select order_purchase_date ,

dateadd(weekday, - datepart(weekday,order_purchase_date)+1,order_purchase_date) as lower,
dateadd(weekday, 7 - datepart(weekday,order_purchase_date),order_purchase_date) as uppper,

format(dateadd(weekday, - datepart(weekday,order_purchase_date)+1,order_purchase_date) , 'dd MMM') + ' to ' +

format(dateadd(weekday, 7 - datepart(weekday,order_purchase_date),order_purchase_date),'dd MMM')

from orders;

Q8 Within 24hours

select * from orders;

We should use order_purchase_date and order_approved_at
difference between both should be 24hours

select count(*)

from orders

where datediff(hour, order_purchase_date, order_approved_at) <= 24;

If we use day instead of hour
select count(*) from orders where datediff(day, order_purchase_date, order_approved_at) <= 1;

not right because we can place today at 6am and it will get delivered at tom 9 pm

Q9. Orders NOT on weekends

select count(*)

from orders

where datepart(weekday, order_estimated_delivery_date) > 1 and

datepart(weekday, order_estimated_delivery_date) < 7;

or we can do using between

```
select count(*)

from orders

where datepart(weekday, order_estimated_delivery_date) between 2 and 6;
```

Q10. CASE statement – Profit or Loss

```
select * from transactions;


select * ,

    case

        when profit > 0 then 'Profit'

        else 'loss'

    end as status

from transactions;
```

Q11. Try by yourself

```
SELECT *,
CASE
  WHEN DATEDIFF(MONTH, joining_date, GETDATE()) >= 12 THEN
'Loyal'
  WHEN DATEDIFF(MONTH, joining_date, GETDATE()) >= 6 THEN
'Potential'
  ELSE 'New'
END AS LoyaltyStatus
FROM mskl_customers;
```